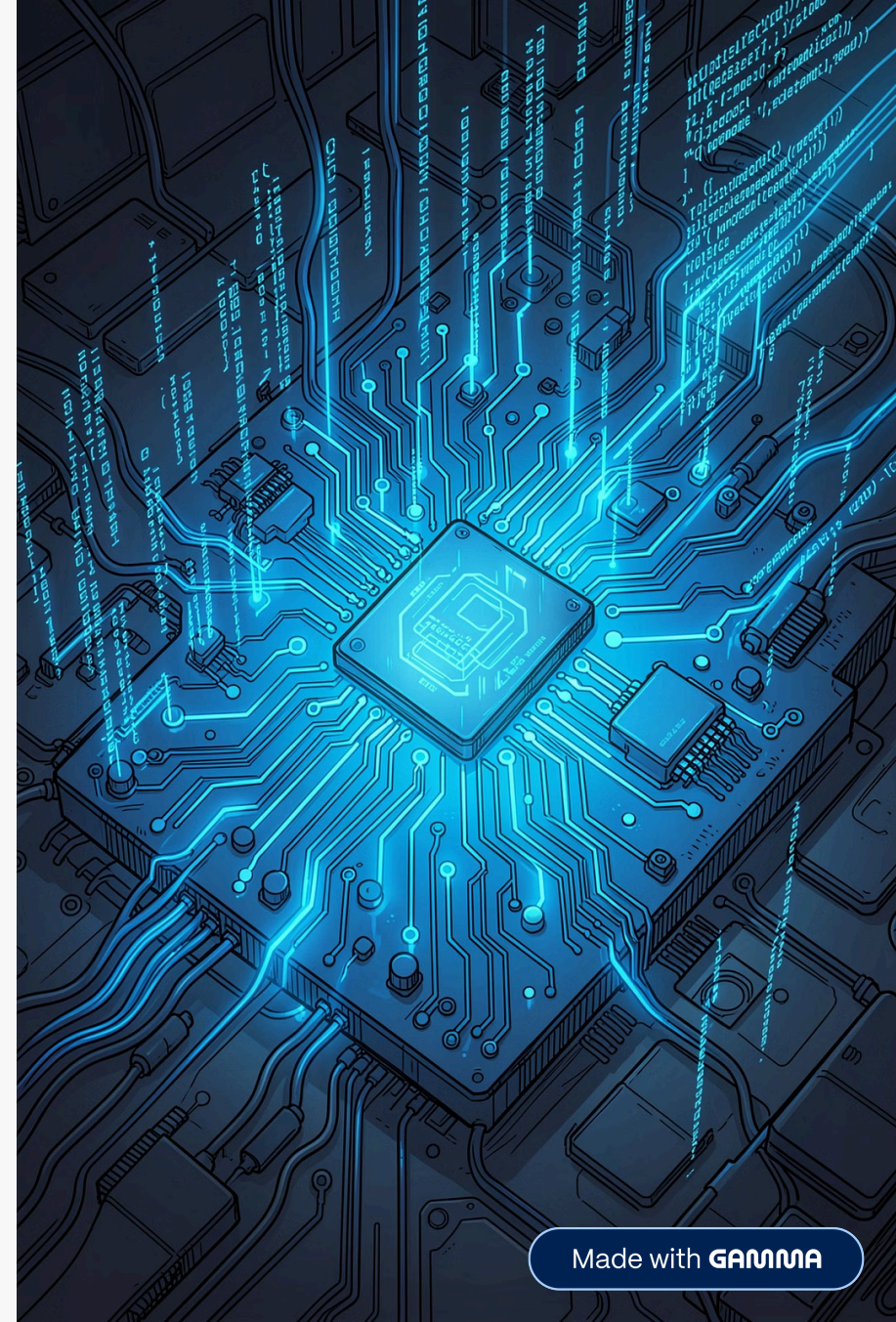


GitHub Repository Analytics Automation Framework

Enterprise Selenium Automation Framework

PRESENTED BY : ANUSHKA SHREE BAJPAI



Project Overview

What Was Built

An enterprise-grade Selenium automation framework that programmatically navigates GitHub Topics, extracts repository analytics, and validates data integrity — all without manual intervention.

The Challenge

Manually auditing hundreds of GitHub repositories is time-consuming and error-prone. This framework automates the entire workflow: discovery, extraction, validation, and reporting.

- ❏ Built to mirror real-world enterprise CI/CD testing pipelines with production-grade patterns.



Technology Stack



Java 11+

Core language with OOP design patterns — Page Object Model, Factory, and Singleton



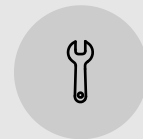
Selenium 4

WebDriver Manager, relative locators, and enhanced Chrome DevTools Protocol integration



TestNG

Flexible test configuration, grouping, parameterization, and built-in assertion libraries



Maven

Dependency management, build lifecycle, and multi-environment profile configuration



Allure Report

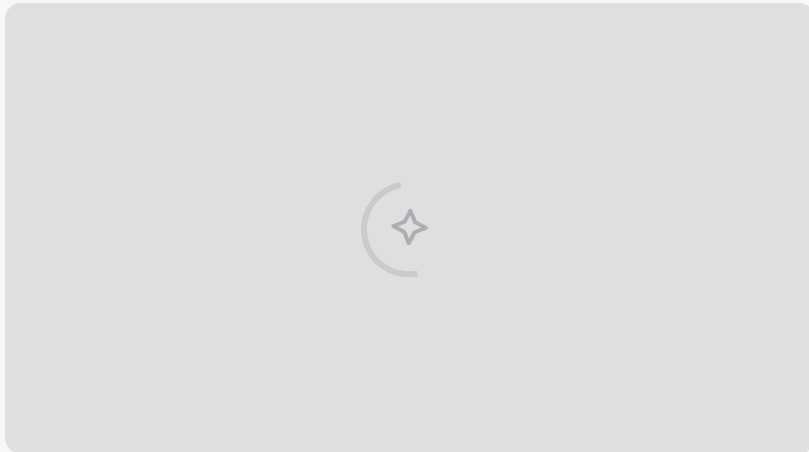
Rich HTML test reports with screenshots, step details, and trend analytics



Log4j2

Asynchronous logging with rolling file appenders and structured log output

Framework Architecture



Layered Design Principles

The framework follows a strict separation of concerns across four distinct layers, ensuring maintainability and scalability.



Test Layer

locators or WebDriver calls



Page Object Layer

Encapsulates UI elements and user interactions



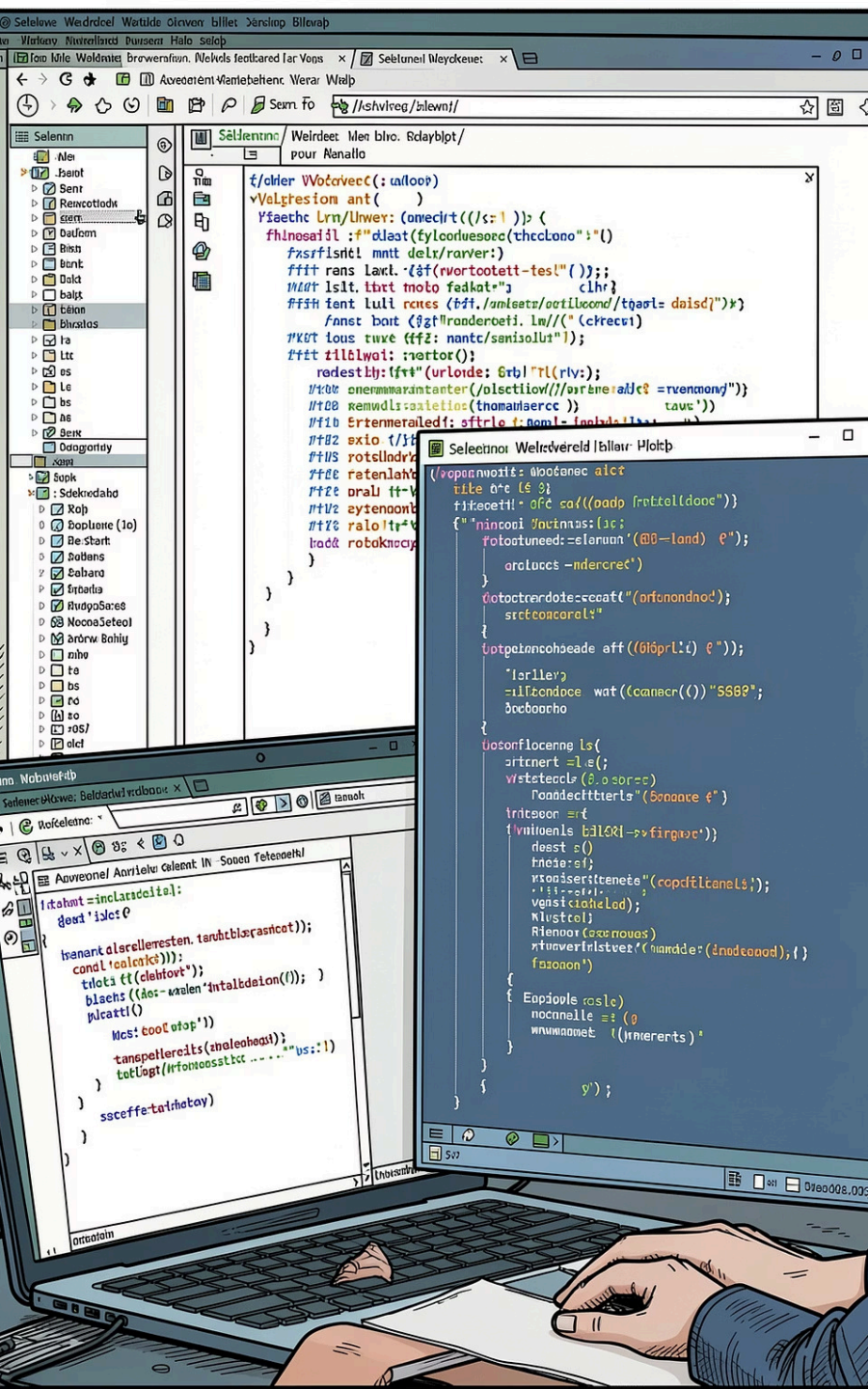
Utility & Infrastructure

Shared helpers, config, logging, and CI integration

Automation Workflow



Each step is encapsulated in a reusable component. The workflow supports headless execution, parallel thread runs, and automatic retry on transient failures — critical for CI pipeline reliability.



Selenium 4 – Core Implementation

WebDriver Manager

Automatically resolves and downloads browser driver binaries – zero manual setup across Windows, Mac, and Linux agents.

Relative Locators

Uses `above()`, `below()`, `near()` for resilient element identification when static IDs are unavailable.

Chrome DevTools Protocol

Intercepts network calls, mocks API responses, and captures performance metrics during test execution.

TestNG Configuration & Parallel Execution

Parallel Test Strategy

Tests execute in parallel across multiple browser threads, dramatically reducing total suite runtime in CI pipelines.

```
<suite name="GitHub Analytics Suite"
  parallel="methods"
  thread-count="4">

  <test name="Chrome Tests">
    <classes>
      <class name="tests.TopicTests"/>
      <class name="tests.RepoTests"/>
    </classes>
  </test>
</suite>
```

`parallel="methods"`

Runs test methods concurrently

`thread-count="4"`

Configurable thread pool size

`data-provider`

Externalized test data via
CSV/JSON

Allure Reporting & Log4j2 Observability

Allure Report Highlights

- Interactive HTML reports with test history and trend charts
- Auto-attached screenshots on failure
- Step-by-step breakdown with annotations
- Severity labels:
`@SeverityLevel.CRITICAL`

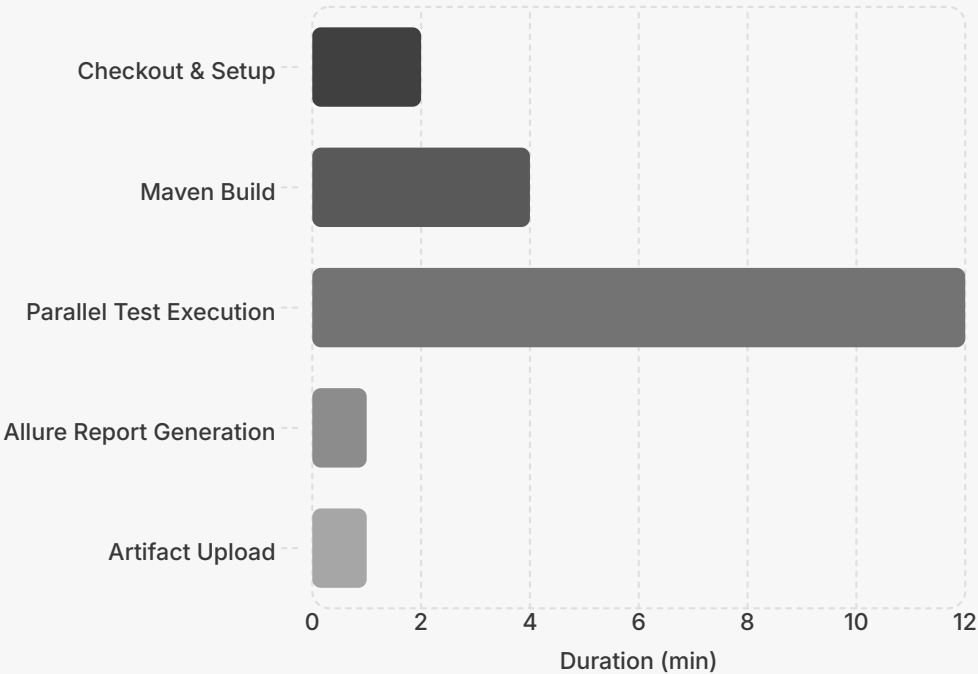
Log4j2 Logging Strategy

- Asynchronous appenders for non-blocking I/O
- Rolling file policy — archives logs by size/date
- Structured JSON log format for ELK stack ingestion



CI/CD Integration & Pipeline Results

Pipeline Stage



Jenkins / GitHub Actions Pipeline

The framework integrates natively with CI pipelines. On every commit, the full test suite executes in a Dockerized environment with results published to Allure and logs archived automatically.

✓ 98% Pass Rate

Across 150+ test cases

⚡ 3.2× Faster

vs. sequential execution

Key Takeaways

1 Enterprise-Grade Patterns

POM, Factory, and Singleton patterns ensure the framework scales cleanly as test coverage grows.

2 Resilience by Design

Explicit waits, retry logic, and headless execution make the suite stable across CI environments.

3 Full Observability

Allure reports + Log4j2 provide end-to-end traceability — from test intent to failure root cause.

4 Production-Ready CI Integration

Maven profiles, Docker support, and GitHub Actions YAML ready for immediate pipeline adoption.



Thank You

ANUSHKA SHREE BAJPAI

Thank you for your time, attention, and support.

- ❏ Questions, feedback, or collaboration ideas? Please feel free to reach out.

